Déborah Boyenval

Postdoctoral Fellow at the University of Helsinki (Finland) – Computational Systems Biology
Ph.D. in Computer Science, Université Côte d'Azur (France)

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Born: 1994 — Nationality: French

Current position

2025–present **Postdoctoral Fellow**, Systems Biology of Drug Resistance in Cancer (Hautaniemi Lab), University of Helsinki. Generative modeling to produce synthetic multi-omics data, and mathematical modeling of tumor evolution to investigate mechanisms underlying treatment resistance.

Academic appointments

2024–2025 **Postdoctoral Fellow**, Laboratoire Bordelais de Recherche en Informatique (LaBRI), Synthesis Team, Bordeaux, France. Mathematical modeling of age-related changes in hematopoietic cell differentiation using multi-omics data from mice.

2018–2022 **Doctoral Teaching Assistant** at Université Côte d'Azur (Computer Science Department) and Polytech Nice Sophia (Engineering School).

Research Interests

Mathematical modeling (continuous, discrete and stochastic) • Generative models (variational autoencoders, transformers) • Multi-omics data analysis • Cancer biology • Causal modeling

Programming Skills

Programming languages: Python (advanced), Java, R, Prolog, MATLAB, SQL, Web.
Deep learning: PyTorch, GPU-accelerated model training with CUDA.
High-performance computing: Experience with the LUMI supercomputer and Slurm-managed computing clusters.
Use of AI: Codex workflows, including agent skills, hooks and developer tools, developement of MCP (Model Context Protocol).
Version control: Git.
Misc: Systems and network administration.

Education

2018–2022 **PH.D. in Computer Science**, Université Côte d'Azur, France (Oct. 2018–Dec. 2022).
Research conducted at I3S (SPARKS Team) and the Institut de Biologie de Valrose.
Thesis entitled "Formal modeling of cyclic biological behaviours with checkpoints: the cell cycle regulation". Supervisors: Gilles Bernot, Jean-Paul Comet and Franck Delaunay.

2016–2018 **MSc in Life Sciences**, Université Côte d'Azur, Sophia Antipolis, France.
Major in **Bioinformatics and Computational Biology**.

Master's thesis (2018) entitled "Mathematical modeling of sex-specific chronotoxicity of an anti-cancer drug: irinotecan". Supervisor: Annabelle Ballesta (currently at Institut Curie, Cancer Systems Pharmacology), Paris.
First-year master's internship (2017): mathematical modeling of circadian clocks, I3S/iBV.

2013–2016 **BSc in Life Sciences**, Université Côte d'Azur, Sophia Antipolis, France.
Third-year specialization in Biology, Informatics and Mathematics (BIM).
Third-year bachelor's internship (2016): wet-lab research in the MINDev Team (Microbiota, Immunity, and Neurodevelopment), Institute of Molecular and Cellular Pharmacology, Nice, France.

Grants, honors & awards

2023 Ph.D. Thesis Award, Fondation Université Côte d'Azur.

Publications

2026 Boyenval, D., Malyutina, A., Oikkonen, J., Liikkanen, S., and Hautaniemi, S. "Synthesis of Ovarian Cancer Multi-Omics Datasets: A Biology-Informed Evaluation Framework". *arXiv preprint, arXiv:2611.08394.* Accepted at the GenAI4Health Workshop at NeurIPS 2026 (oral presentation) and in the Findings track of ML4H 2026. [PDF]

2025 Leppiniemi, S., Afenteva, D., Li, Y., Micoli, G., Lavikka, K., Holmström, S., Boyenval, D., Oikkonen, J., Hautaniemi, S., and Muranen, T. A. "SegmentQTL: Identifying genetic variants influencing molecular phenotypes in copy number-driven cancers". *bioRxiv.* [DOI] [PDF]

2024 Chevalier, S., Boyenval, D., Magaña-Lopez, G., Roncalli, T., Vaginay, A., and Paulevé, L. "BoNesis: a Python-based declarative environment for the verification, reprogramming, and synthesis of Most Permissive Boolean networks". *Computational Methods in Systems Biology (CMSB 2024).* [DOI]

2022 Dulong, S., Souza, L., Machowiak, J., Peuteman, B., Duvallet, G., Boyenval, D., Roth, E., Asgarova, A., Chang, Y., Li, X.-M., Foudi, A., and Ballesta, A. "Sex and Circadian Timing Modulate Oxaliplatin Hematological and Hematopoietic Toxicities". *Pharmaceutics*, 14, 2465. [DOI]

2020 Boyenval, D., Bernot, G., Collavizza, H., and Comet, J.-P. "What is a cell cycle checkpoint? The ToTemBioNet answer". *Computational Methods in Systems Biology (CMSB 2020).* [DOI]

Doctoral thesis

2022 Boyenval, D. "Formal modeling of cyclic biological behaviours with checkpoints: the cell cycle regulation". *Ph.D. thesis in Computer Science.*
Université Côte d'Azur; Laboratoire d'Informatique Signaux et Systèmes de Sophia Antipolis (I3S) and Institut de Biologie de Valrose, Nice, France.
Defended December 15, 2022. Supervisors: Gilles Bernot, Jean-Paul Comet, and Franck Delaunay. [HAL]

Posters and Talks

Talks

Dec 2026 "Synthesis of Ovarian Cancer Multi-Omics Datasets: A Biology-Informed Evaluation Framework". GenAI4Health Workshop at NeurIPS 2026 (Sydney). *Upcoming oral presentation.*

2023 "Formal modeling of the discrete dynamics of intrinsic cell cycle checkpoints: a proof of concept". GT-Bioss (Online).

Sep 2022 Talk on the principles of model checking and a practical application in biology for Women in Machine Learning (WiML). [SLIDES] (Jussieu, Paris).

2022 "A discrete modeling study devoted to the formalization and verification of mammalian cell cycle checkpoints". Lifeware Public Seminar (Inria Saclay, France).

2021 "Mammalian cell cycle, formalizing phases". GT-Bioss Annual Days. [SLIDES] (Lyon, France).

Posters

Dec 2026 "Synthesis of Ovarian Cancer Multi-Omics Datasets: A Biology-Informed Evaluation Framework". ML4H 2026 Symposium (Sydney). *Upcoming poster presentation.*

2026 "Generating Synthetic Data from Ovarian Cancer Patients". RPU Science Day 2026 (Helsinki).

2025 "Learning Resistance: Synthetic Data for Predictive Modeling in HGSC", DECIDER Annual Meeting (Helsinki).

2021 "Logical and incremental formalization of cell cycle checkpoints". JOBIM 2021, July (held online due to COVID-19).

2019 "Introduction of Priorities in Biological Regulatory Networks". Modelife Annual Meeting (Fréjus, France).

2019 "A discrete cell cycle model: from phase characterization toward observable properties verification". Summer School on Formal Modeling of Biological Regulatory Networks (Porquerolles, France).

Teaching

2025–2026 Contributed to the *Introduction to Systems Biology* course (Spring 2025).
Teaching relational databases as part of the *Clinical Data Mining* course (Fall 2026). Both courses are coordinated by Sampsa Hautaniemi at the University of Helsinki.

2018–2021 Teaching Assistant, Université Côte d'Azur (BSc and MSc levels).
Web development, systems and network administration, database theory and SQL, imperative programming and Python, algorithms for biology.

Service

Reviewer for PLOS Computational Biology.

2019–2021 Treasurer, ADAMS Doctoral Students' Association, Université Côte d'Azur.

2018–2019 FabLab Manager, Valrose campus of Université Côte d'Azur.

Referees

Sampsa Hautaniemi (Postdoctoral referee) — sampsa.hautaniemi@helsinki.fi; +358 29 412 5419.
Jean-Paul Comet (Ph.D. referee) — jean-paul.comet@univ-cotedazur.fr; +33 4 89 15 43 49.
Gilles Bernot (Ph.D. referee) — gilles.bernot@univ-cotedazur.fr; +33 4 89 15 43 84.

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